Information Theory Workshop (ITW) 2026

Information-theoretic foundations of anytime-valid inference

Part 2 (11:00 am to 12:30pm)

Shubhanshu Shekhar
University of Michigan, Ann Arbor
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Tutorial outline

PART 1:

Introduction + finite alphabet methods

1. Motivation

A/B testing example; failure of classical methods; Anytime-valid (AV) testing

2. Fundamental Limits

Classical DPI + Wald arguments

3. AV Test for Finite Alphabets

Universal Inference; Proof of optimality

PART 2:

General Principles

1. Bounded mean testing

Testing-by-betting principle, Kelly criterion, universal portfolios.

2. Unifying Design Principle

Donsker-Varadhan representation, Likelihood-free Universal Inference.

3. Extensions

Sequential estimation, change detection, non-iid data, etc.

Recap: the sequential testing problem

- Sequential testing problem on $\mathcal{X}$:

$$X_1, X_2, \ldots \overset{\text{iid}}{\sim} P.$$

$$H_0 : P_X \in \mathcal{P}_0 \qquad \text{vs.} \qquad H_1 : P_X \in \mathcal{P}_1.$$

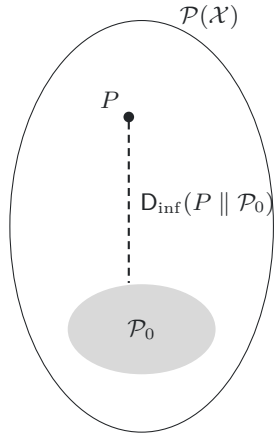
- Design a stopping time τ_α at which we reject H_0, with

$$\underbrace{\sup_{Q \in \mathcal{P}_0} \mathbb{P}_Q(\tau_\alpha < \infty) \leq \alpha,}_{\text{Validity under } H_0} \qquad \underbrace{\inf_{P \in \mathcal{P}_1} \mathbb{P}_P(\tau_\alpha < \infty) = 1\,.}_{\text{“Power-one” under } H_1}$$

- Any such τ_α must satisfy

$$\mathbb{E}_P[\tau_\alpha] \geq \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)},$$

where $\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0) := \inf_{Q \in \mathcal{P}_0} D(P\|Q)$.



Goal: Design τ_α that matches this limit as $\alpha \downarrow 0$.

Recap: universal inference

- In Part 1, we focused on $\mathcal{X} = \{1, \ldots, k\}$.

- **General approach:**
  (i) Define a process $\{E_t : t \geq 1\}$ that quantifies evidence against H_0.
  (ii) Reject H_0 when E_t reaches the threshold $1/\alpha$.

- In Part 1, we used a mixture-to-maximum likelihood ratio process:

$$E_t = \frac{\displaystyle\int_{\Delta_{k-1}} \prod_{i=1}^{t} q(X_i)\, \pi(dq)}{\displaystyle\sup_{Q \in \mathcal{P}_0} \prod_{i=1}^{t} Q(X_i)} = \frac{\text{mixture likelihood}}{\text{maximum null likelihood}}.$$

- However, in many problems, a full likelihood is unavailable or difficult to compute.

A nonparametric example: bounded-mean testing

- **Observations:** $\mathcal{X} = [0,1]$, and $X_1, X_2, \ldots \overset{\text{iid}}{\sim} P$
- **Testing problem:** For some fixed $m \in (0,1)$, decide b/w

$$H_0 : \mathbb{E}_P[X] \leq m, \quad \text{versus} \quad H_0 : \mathbb{E}_P[X] > m.$$

- The null posits that the mean of this stream equals m; equivalently,

$$\mathcal{P}_0 = \big\{P \in \mathcal{P}([0,1]) : \mathbb{E}_P[X] \leq m\big\}.$$

- **There is no common σ-finite dominating measure** for $\mathcal{P}_0$.
- The same is true for many other problems; e.g., *two-sample testing, independence testing etc.*

The principle of testing by betting

Evidence against a null hypothesis can be formally quantified as the gain in wealth of a fictitious bettor repeatedly betting on observed outcomes, at odds that are fair or subfair under H_0.

Initialize: $W_0 \leftarrow 1$.

For $t = 1, 2, \ldots$ **do:**

1. **Place a bet:** Choose a fraction $u_t \in [0, 1]$ of current wealth to bet, and the payoff rule.
2. **Observe** the next outcome X_t, and collect the payoff.
3. **Update** the bettor's wealth W_t.

End for.

Designing a (sub-)fair bet for $H_0 : \mathbb{E}[X_t] \leq m$

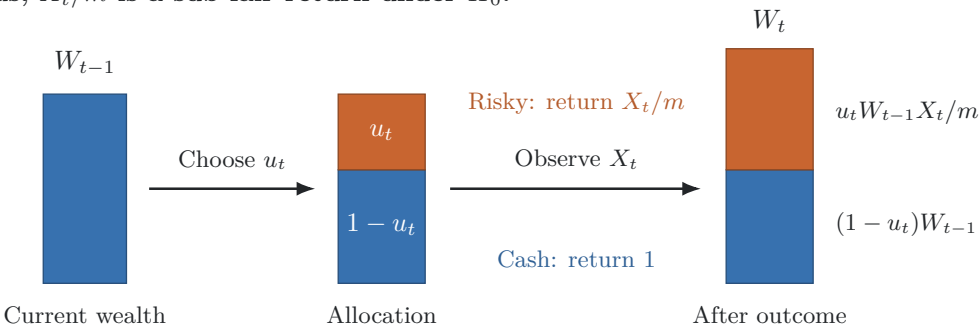
$$\mathbb{E}_Q[X_t] \leq m \quad \Longrightarrow \quad \mathbb{E}_Q\left[\frac{X_t}{m}\right] \leq 1, \quad \text{for all } Q \in \mathcal{P}_0.$$

Thus, X_t/m is a sub-fair return under H_0.

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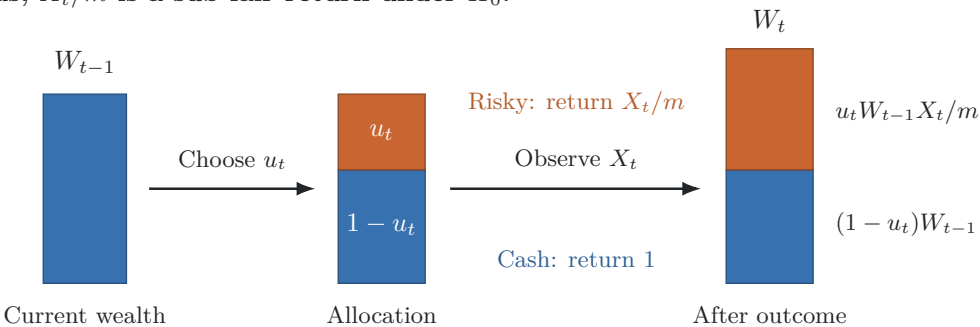
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Thus, X_t/m is a sub-fair return under H_0.



$$W_t = W_{t-1}\left((1-u_t) + u_t\frac{X_t}{m}\right), \qquad 0 \le u_t \le 1.$$

Testing by betting for bounded means

Testing $H_0 : \mathbb{E}[X] \leq m$, with $X_t \in [0,1]$.

Initialize: $W_0 = 1$.

For $t = 1, 2, \ldots$:

1. **Choose** $u_t \in [0,1]$ using only X^{t-1}.
2. **Observe** X_t.
3. **Update** $W_t = W_{t-1}\big(1 - u_t + u_t X_t / m\big)$.
4. **Reject** H_0 and stop if $W_t \geq 1/\alpha$.

$$\tau_\alpha = \inf\{t \geq 1 : W_t \geq 1/\alpha\}$$

How should we choose the sequence of bets $\{u_t\}$?

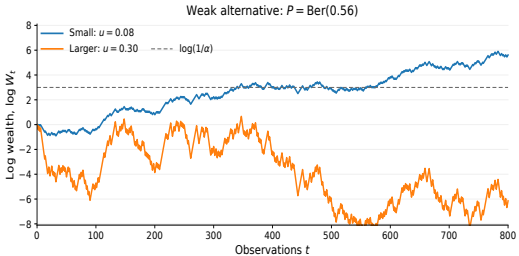
Testing by betting for bounded means

Under H_0**:** $X_t \sim \mathrm{Ber}(0.50)$. Ten independent runs.

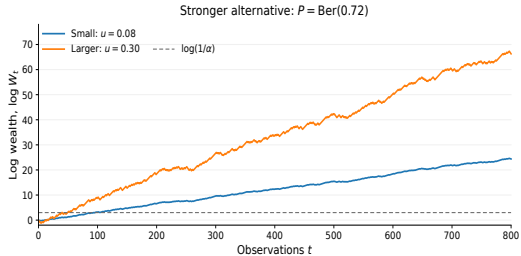


Under the null, no bet can gain money consistently.

Testing by betting for bounded means



The smaller bet works well.



The larger bet works well.

Under the alternative, different bets might perform better.

Any betting strategy works under H_0

Recall that $u_t \in [0,1]$ is chosen using only $\mathcal{F}_{t-1} = \sigma(X_1, \ldots, X_{t-1})$, and

$$W_t = W_{t-1}\left(1 - u_t + u_t \frac{X_t}{m}\right), \qquad W_0 = 1.$$

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$$W_t = W_{t-1}\left(1 - u_t + u_t \frac{X_t}{m}\right), \qquad W_0 = 1.$$

Under any $Q \in \mathcal{P}_0$,

$$\mathbb{E}_Q[W_t \mid \mathcal{F}_{t-1}] = W_{t-1}\left(1 - u_t + u_t \frac{\mathbb{E}_Q[X_t]}{m}\right) \leq W_{t-1}.$$

Thus, $\{W_t\}_{t \geq 0}$ is a **nonnegative supermartingale** under H_0.

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Thus, $\{W_t\}_{t\ge 0}$ is a **nonnegative supermartingale** under H_0.

By Ville's inequality, we have

$$\sup_{Q \in \mathcal{P}_{0,m}} \mathbb{P}_Q\left(\sup_{t\ge 0} W_t \ge \frac{1}{\alpha}\right) \le \alpha, \quad \text{hence} \quad \tau_\alpha = \inf\left\{t \ge 1 : W_t \ge \frac{1}{\alpha}\right\} \text{ is valid for } \mathcal{P}_0$$

Ville. *Étude critique de la notion de collectif* (1939)

How should we bet?

Suppose we use a fixed $u \in [0, 1]$ in every round:

$$W_n(u) = \prod_{t=1}^{n} \left(1 - u + u\frac{X_t}{m}\right).$$

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Under an i.i.d. alternative P, by the SLLN,

$$\begin{aligned}\frac{1}{n}\log W_n(u) &= \frac{1}{n}\sum_{t=1}^{n} \log\left(1 - u + u\frac{X_t}{m}\right) \\ &\xrightarrow{\text{a.s.}} \underbrace{\mathbb{E}_P\left[\log\left(1 - u + u\frac{X}{m}\right)\right]}_{=:g_P(u)}.\end{aligned}$$

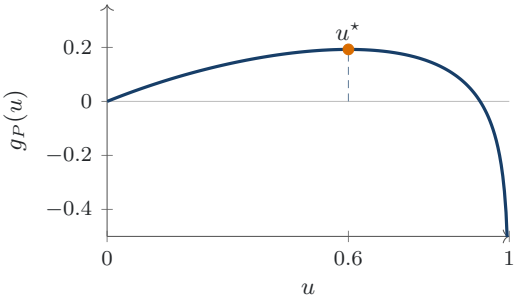
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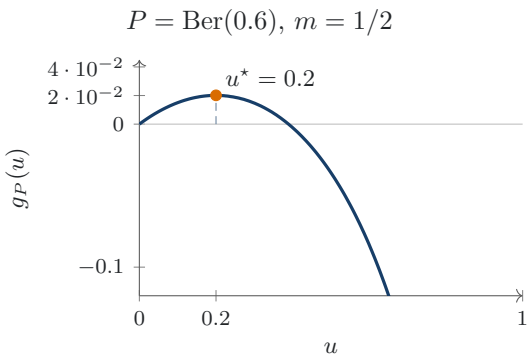


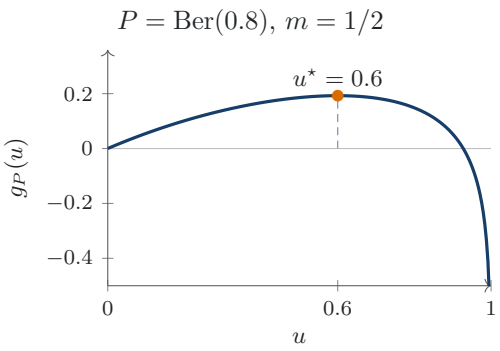
The **log-optimal or Kelly bet** is

$$u^\star \in \arg\max_{u \in [0,1]} g_P(u).$$

Learning the Kelly bet by mixing

Challenge. The Kelly bet $u^\star$ depends on the unknown alternative P.





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Idea. Instead of committing to a single u, mix over all constant allocations $u \in [0,1]$.

$$\overline{W}_t = \int_0^1 W_t(u)\,\pi(du), \qquad W_t(u) = \prod_{i=1}^{t} \left(1 - u + u\frac{X_i}{m}\right).$$

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Validity. Mixing preserves the nonnegative supermartingale property under every $Q \in \mathcal{P}_0$.

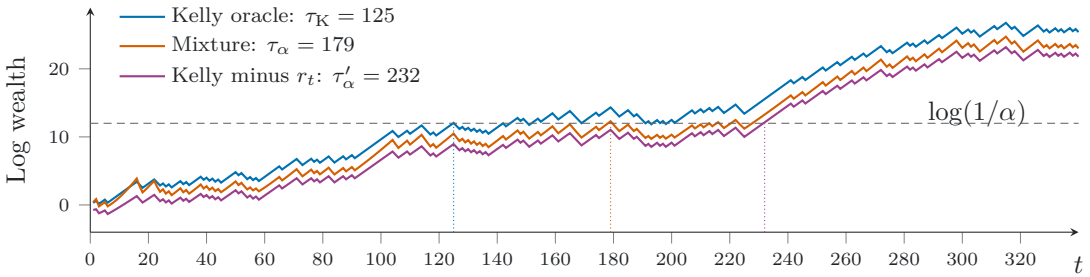
Efficiency? How much log-wealth does the mixture lose relative to the Kelly bet $u^\star$?

From portfolio regret to stopping time

For mixing distribution $\pi = \text{Beta}\left(\frac{1}{2}, \frac{1}{2}\right)$, we have a pathwise guarantee

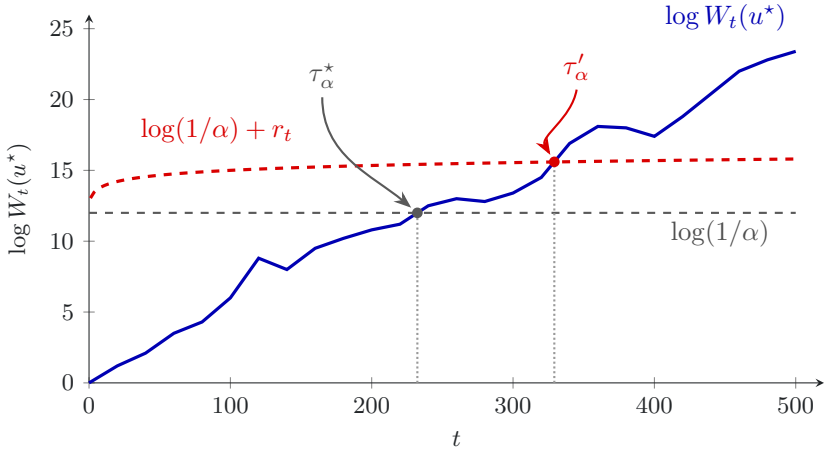
$$\log \overline{W}_t \geq \sup_{u \in [0,1]} \log W_t(u) - r_t, \qquad r_t = \tfrac{1}{2}\log(t+1) + \log 2.$$

Hence, $\tau_\alpha \leq \tau'_\alpha := \inf\{t \geq 1 : \log W_t(u^\star) - r_t \geq \log(1/\alpha)\}$.



From portfolio regret to stopping time

$$\tau'_\alpha = \inf\left\{t \geq 1 : \log W_t(u^\star) \geq \log(1/\alpha) + r_t\right\}, \qquad r_t = \frac{1}{2}\log(t+1) + \log 2.$$



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By the regret bound, we have $\tau_\alpha \leq \tau'_\alpha$.

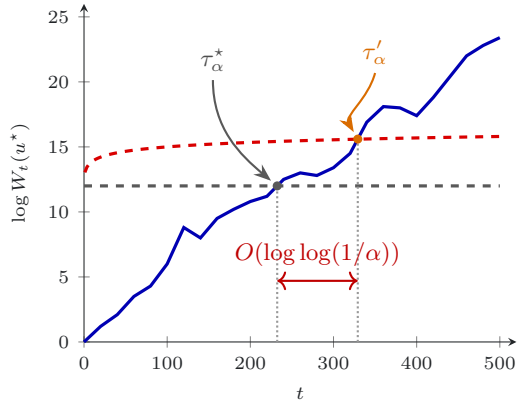
Under P, the oracle process grows linearly:

$$\frac{1}{t}\log W_t(u^\star) \to g_P^\star > 0.$$

Hence, τ_α is finite a.s. (**power one**).

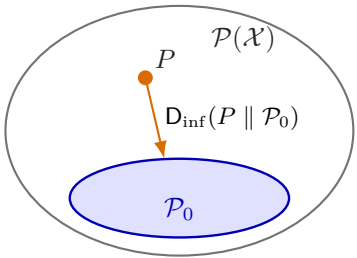
Since $\tau_\alpha \leq \tau'_\alpha$, we also have

$$\mathbb{E}_P[\tau_\alpha] \leq \frac{\log(1/\alpha) + O(\log\log(1/\alpha))}{g_P^\star}.$$



Is this test optimal?

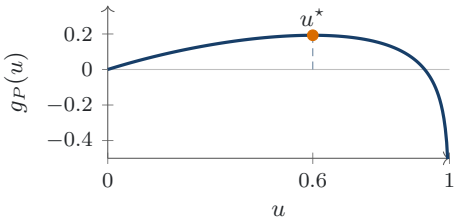
Fundamental limit (Part 1)



Every level-α, power-one test satisfies

$$\mathbb{E}_P[\tau_\alpha] \geq \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_{0,m})}.$$

Testing-by-betting

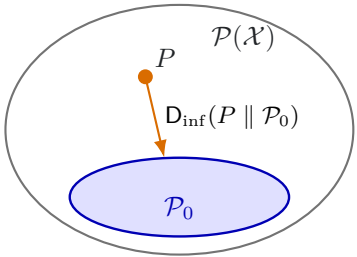


The proposed test τ_α achieves

$$\mathbb{E}_P[\tau_\alpha] \leq \frac{\log(1/\alpha)}{g_P^\star}\,(1 + o(1))\,.$$

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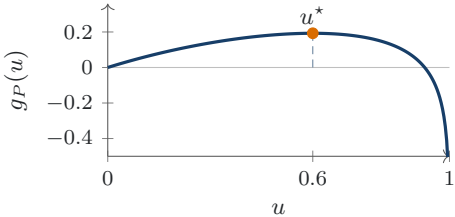
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Testing-by-betting



The proposed test τ_α achieves

$$\mathbb{E}_P[\tau_\alpha] \leq \frac{\log(1/\alpha)}{g_P^\star}\,(1 + o(1))\,.$$

Is the Kelly growth rate equal to the minimum relative entropy?

$$g_P^\star \stackrel{?}{=} \mathsf{D}_{\inf}(P \parallel \mathcal{P}_{0,m}).$$

Kelly growth-rate is dual $\mathsf{D}_{\inf}$

For P on $[0,1]$ with $\mathbb{E}_P[X] > m$, recall

$$\mathsf{D}_{\inf}(P \parallel \mathcal{P}_{0,m}) = \inf_{\substack{Q \in \mathcal{P}([0,1]) \\ \mathbb{E}_Q[X] \le m}} D(P \parallel Q).$$

This is an **infinite-dimensional convex program**, but with one inequality constraint

It turns out that its dual is precisely the **Kelly growth-rate maximization**:

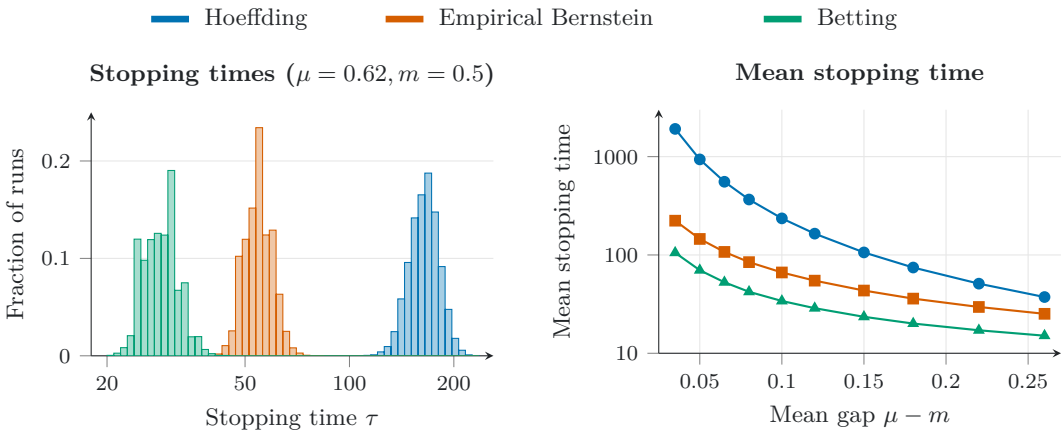
$$\mathsf{D}_{\inf}(P \parallel \mathcal{P}_{0,m}) = \sup_{u \in [0,1]} \mathbb{E}_P \left[\log \left(1 - u + u \frac{X}{m} \right) \right] := g_P^\star.$$

Combining this duality with our stopping-time bounds,

$$\lim_{\alpha \downarrow 0} \frac{\mathbb{E}_P[\tau_\alpha]}{\log(1/\alpha)} = \frac{1}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_{0,m})}.$$

Empirical Comparison

Baselines: Tests based on time-uniform concentration inequalities

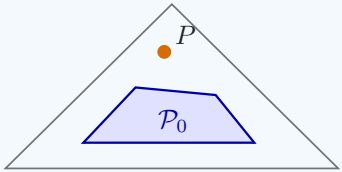


Summary of the two approaches

Finite alphabets

Likelihood-based universal inference

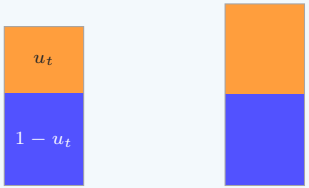
$$E_t = \frac{\text{KT predictor}}{\text{maximum null likelihood}}$$



Bounded-mean testing

Testing by betting

$$W_t = \prod_{i=1}^{t} \left(1 - u_i + u_i \frac{X_i}{m}\right)$$



Is there a common design principle for a general testing problem?

A Unifying Design Principle

Likelihood-free Universal Inference

Testing over general observation spaces

For some Polish space $\mathcal{X}$, let

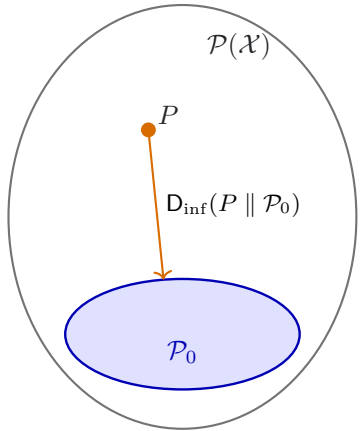
$$X_1, X_2, \ldots \overset{\text{iid}}{\sim} P, \qquad P \in \mathcal{P}(\mathcal{X}) \text{ unknown.}$$

Given $\mathcal{P}_0 \subseteq \mathcal{P}(\mathcal{X})$, we wish to test

$$H_0 : P \in \mathcal{P}_0 \qquad \text{vs.} \qquad H_1 : P \notin \mathcal{P}_0.$$

Lower Bound: For any level-α, power-one test,

$$\mathbb{E}_P[\tau_\alpha] \geq \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)}, \quad \mathsf{D}_{\inf}(P \parallel \mathcal{P}_0) \coloneqq \inf_{Q \in \mathcal{P}_0} D(P\|Q).$$



Can we construct tests attaining this limit without relying on likelihood functions?

Donsker–Varadhan (DV) Representation

For any $P, Q \in \mathcal{P}(\mathcal{X})$, recall the **Donsker–Varadhan variational representation**:

$$\mathsf{D}(P \parallel Q) = \sup_{f \in C_b(\mathcal{X})} \left\{ \mathbb{E}_P[f(X)] - \log \mathbb{E}_Q[e^{f(X)}] \right\},$$

where $C_b(\mathcal{X})$ denotes the space of bounded continuous functions on $\mathcal{X}$.

Interestingly, when $P \ll Q$ and $\log(dP/dQ) \in C_b(\mathcal{X})$, an optimizer is

$$f_Q^\star(x) = \log \frac{dP}{dQ}(x).$$

Key observation: The DV representation allows us to write the log-likelihood ratio as the solution of an optimization problem.

Likelihood-Free UI for $\mathcal{P}_0 = \{Q\}$

For any predictable sequence $f_t \in C_b(\mathcal{X})$, define

$$M_0^Q = 1, \qquad M_t^Q = \prod_{i=1}^{t} \frac{e^{f_i(X_i)}}{\mathbb{E}_Q[e^{f_i(X)}]}.$$

$$X^{t-1} \xrightarrow{\text{Choose}} f_t \xrightarrow{\text{Observe}} X_t \xrightarrow{\text{Update}} M_t^Q$$

Validity. $\{M_t^Q\}$ is a nonnegative martingale under Q:

$$\mathbb{E}_Q[M_t^Q \mid \mathcal{F}_{t-1}] = M_{t-1}^Q \frac{\mathbb{E}_Q[e^{f_t(X)}]}{\mathbb{E}_Q[e^{f_t(X)}]} = M_{t-1}^Q.$$

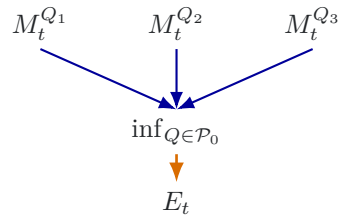
Log-growth. For a fixed function $f_t \equiv f$, under an i.i.d. alternative P,

$$\frac{1}{t} \log M_t^Q \quad = \quad \frac{1}{t} \sum_{i=1}^{t} f_t - \log \mathbb{E}_Q[e^{f_t}] \quad \xrightarrow{\text{a.s.}} \quad \mathbb{E}_P f - \log \mathbb{E}_Q e^f.$$

Likelihood-Free UI for Composite $\mathcal{P}_0$

For each $Q \in \mathcal{P}_0$, construct $\{M_t^Q\}$ using predictable $\{f_{t,Q}\}$

$$M_t^Q = \prod_{i=1}^{t} \frac{e^{f_{i,Q}(X_i)}}{\mathbb{E}_Q[e^{f_{i,Q}(X)}]}.$$



As in Part 1, take the pointwise infimum:

$$E_t = \inf_{Q \in \mathcal{P}_0} M_t^Q.$$

Validity. $\tau_\alpha = \inf\{t \geq 1 : E_t \geq 1/\alpha\}$ satisfies uniform level-α property under $\mathcal{P}_0$.

$E_t \leq M_t^{Q_0}$ Ville's ineq.

$$\sup_{Q_0 \in \mathcal{P}_0} \mathbb{P}_{Q_0}\left(\sup_{t \geq 0} E_t \geq \frac{1}{\alpha}\right) \leq \sup_{Q_0 \in \mathcal{P}_0} \mathbb{P}_{Q_0}\left(\sup_{t \geq 0} M_t^{Q_0} \geq \frac{1}{\alpha}\right) \leq \alpha.$$

Summary of the Likelihood-Free UI Algorithm

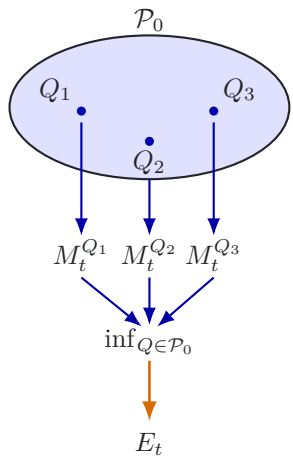
Initialize: $M_0^Q = 1$ for every $Q \in \mathcal{P}_0$.

For $t = 1, 2, \ldots$:

1. **Choose** predictable $f_{t,Q}$ for each $Q \in \mathcal{P}_0$.
2. **Observe** the next X_t.
3. **Update** each null-specific martingale

$$M_t^Q = M_{t-1}^Q \frac{e^{f_{t,Q}(X_t)}}{\mathbb{E}_Q[e^{f_{t,Q}(X)}]}.$$

4. **Aggregate.** $E_t = \inf_{Q \in \mathcal{P}_0} M_t^Q$.
5. **Reject** H_0 if $E_t \geq 1/\alpha$.

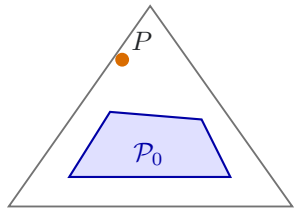


Recovering finite-alphabet universal inference

Suppose $\mathcal{X} = \{1, \ldots, k\}$, and assume each null pmf q is positive.

Predictable function: For each $t \geq 1$, choose

$$f_{t,Q}(x) = \log \frac{\widehat{p}_t(x)}{q(x)}, \quad \text{where} \quad \widehat{p}_t = \text{KT-predictor}$$



Null normalization. For every Q with pmf q, we have

$$\mathbb{E}_Q[e^{f_{t,Q}(X)}] = \sum_{x \in \mathcal{X}} q(x) \frac{\widehat{p}_t(x)}{q(x)} = 1.$$

Substituting into likelihood-free UI gives

$$E_t = \inf_{Q \in \mathcal{P}_0} \prod_{i=1}^{t} \frac{\widehat{p}_i(X_i)}{q(X_i)} = \frac{\widetilde{P}(X^t)}{\sup_{Q \in \mathcal{P}_0} Q^t(X^t)}.$$

Recovering bounded-mean betting

Let $X_t \in [0,1]$ and $\mathcal{P}_0 = \{Q : \mathbb{E}_Q X \le m\}$.

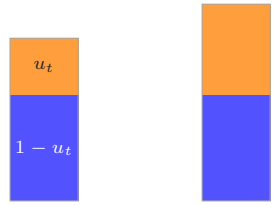
Predictable function: $f_{t,Q} = f_{u_t}$ for predictable u_t, where

$$f_u(x) = \log\left(1 - u + u\frac{x}{m}\right), 0 \le u < 1.$$

Null normalization. For every Q with $\mathbb{E}_Q[X] \le m$,

$$\mathbb{E}_Q[e^{f_{t,Q}(X)}] = 1 - u_t + u_t\frac{\mathbb{E}_Q X}{m} \le 1,$$

with equality if $\mathbb{E}_Q X = m$ (e.g., $Q = \text{Bernoulli}(m)$).



On taking an infimum over Q, we have

$$E_t = \inf_{Q \in \mathcal{P}_0} M_t^Q = \prod_{i=1}^{t}\left(1 - u_i + u_i\frac{X_i}{m}\right) = W_t.$$

Beyond the two examples

Two-sample testing

Observe independent $X_t \sim P_X$ and $Y_t \sim P_Y$.

$$H_0 : \quad P_X = P_Y.$$

Independence testing

Observe i.i.d. pairs $(X_t, Y_t) \sim P_{XY}$.

$$H_0 : \quad P_{XY} = P_X \otimes P_Y.$$

Invariance testing

Observe $X_t \sim P_X$; let G act on $\mathcal{X}$.

$$H_0 : \quad X \overset{d}{=} g(X), \quad \forall g \in G.$$

Log-concavity testing

Observe i.i.d. $X_t \sim P$ on $\mathbb{R}^d$, with density p.

$$H_0 : \quad p \text{ is log-concave.}$$

The same likelihood-free UI idea applies to all these problems.

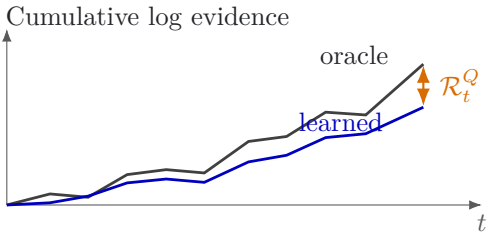
The price of learning: regret

Fix an alternative $P \notin \mathcal{P}_0$. For each $Q \in \mathcal{P}_0$, compare:

- the **oracle** log likelihood process: $\Lambda_t^Q = \sum_{i=1}^t \log \frac{dP}{dQ}(X_i)$,
- the **learned** log evidence $\log M_t^Q$.

Regret: The gap between oracle and learned log evidence,

$$\mathcal{R}_t^Q := \Lambda_t^Q - \log M_t^Q.$$



KT prediction: regret w.r.t. to a fixed distribution.

Universal portfolios: regret w.r.t. to a fixed bet.

Stronger regret control $\Longrightarrow$ stronger guarantees

Fix an alternative P with $\mathsf{D}_{\inf}(P \,\|\, \mathcal{P}_0) = \inf_{Q \in \mathcal{P}_0} \mathsf{D}(P \,\|\, Q) > 0$.

Let $\mathcal{R}_t = \sup_{Q \in \mathcal{P}_0} \mathcal{R}_t^Q$ be the *uniform regret* over the null distribution class.

Control on $\mathcal{R}_t$	**Statistical Guarantee**	
$\limsup_{t \to \infty} \dfrac{\mathcal{R}_t}{t} < \mathsf{D}_{\inf}(P \,\|\, \mathcal{P}_0)$ *a.s.*	$\mathbb{P}_P(\tau_\alpha < \infty) = 1$	(Power one)
$\mathcal{R}_t \le \rho t$, $\rho < \mathsf{D}_{\inf}(P \,\|\, \mathcal{P}_0)$, *w.h.p.*	$\mathbb{E}_P[\tau_\alpha] < \infty$	(Finite expectation)
$\mathcal{R}_t \le r_t$ with $r_t = o(t)$, *w.h.p.*	$\mathbb{E}_P[\tau_\alpha] \sim \dfrac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P \,\|\, \mathcal{P}_0)}$	(First-order opt.)

Sufficient conditions for power one property

Under certain conditions, we can show that

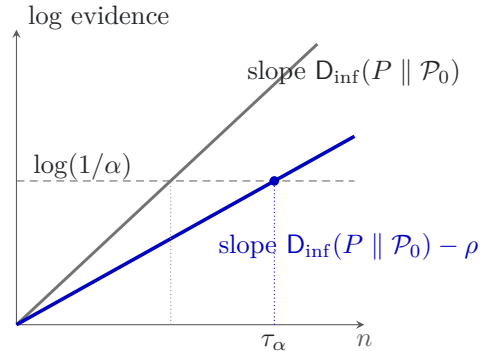
$$\frac{\log E_t}{t} \gtrsim \mathsf{D}_{\inf}(P \parallel \mathcal{P}_0) - \frac{\mathcal{R}_t}{t}.$$

Suppose $\limsup_{t\to\infty} \frac{\mathcal{R}_t}{t} \leq \rho < \mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)$ a.s.

Then, the asymptotic growth rate is positive:

$$\liminf_{t\to\infty} \frac{\log E_t}{t} \geq \mathsf{D}_{\inf}(P \parallel \mathcal{P}_0) - \rho > 0.$$

Hence, $\{E_t : t \geq 1\}$ crosses $\log(1/\alpha)$ w.p. 1.

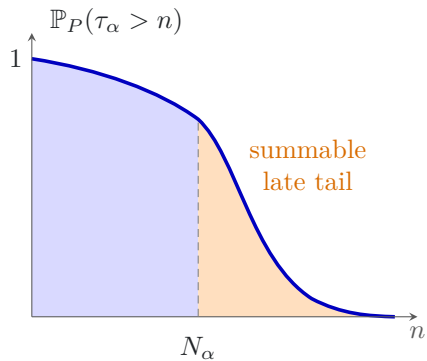


From regret to expected stopping time

$$\mathbb{E}_P[\tau_\alpha] = \sum_{n\geq 0} \mathbb{P}_P(\tau_\alpha > n) \leq N_\alpha + \underbrace{\sum_{n\geq N_\alpha} \mathbb{P}_P(\tau_\alpha > n)}_{\text{Late-stopping tail}}.$$

Finite mean: A summable late-stopping tail for some N_α is sufficient.

Optimality: Choose $N_\alpha = \left\lceil \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P\|\mathcal{P}_0)-\varepsilon} \right\rceil$.



Under stronger regret control, can show

$$\mathbb{E}_P[\tau_\alpha] \leq N_\alpha + O_{P,\varepsilon}(1).$$

Letting $\alpha \downarrow 0$ and then $\varepsilon \downarrow 0$:

$$\lim_{\alpha\downarrow 0} \frac{\mathbb{E}_P[\tau_\alpha]}{\log(1/\alpha)} = \frac{1}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)}.$$

Summary of the general design principle

1. **Construct null-specific martingales.**
   For every $Q \in \mathcal{P}_0$, construct a nonnegative martingale $\{M_t^Q\}_{t\geq 0}$ under Q, using predictable witness functions $\{f_{t,Q} : t \geq 1\}$.

2. **Aggregate by pointwise minimization.**

$$E_t = \inf_{Q\in\mathcal{P}_0} M_t^Q.$$

   The resulting e-process is anytime-valid under every $Q \in \mathcal{P}_0$.

3. **Control regret to guarantee efficiency.**
   Stronger regret and tail guarantees yield stronger statistical performance:

$$\text{Power one} \quad \longrightarrow \quad \text{Finite } \mathbb{E}_P[\tau_\alpha] \quad \longrightarrow \quad \text{First-order optimality.}$$

Some Extensions Beyond I.I.D. Testing

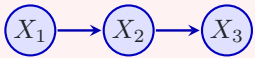
Beyond the basic i.i.d. testing problem

Our basic i.i.d. testing framework has three ingredients:

Observation model	**Inferential task**	**Performance criterion**
$X_t \overset{\text{i.i.d.}}{\sim} P$	$H_0 : \ P \in \mathcal{P}_0$	$\mathbb{E}_P[\tau_\alpha]$
Fully observed, fixed P	Reject or continue	Stop quickly on average

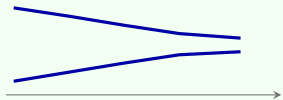
Change the observations

Markov chains; censored data



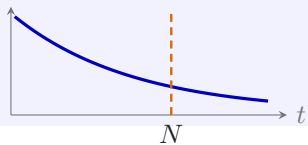
Change the task

Confidence sequences; change detection



Change the objective

Deadlines; time-sensitive rewards



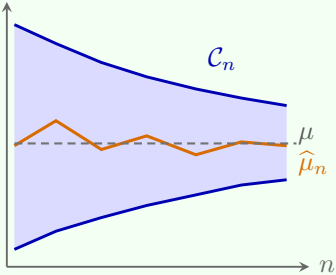
Modified Task 1: Estimation

Suppose $X_1, X_2, \ldots \overset{\text{i.i.d.}}{\sim} P$, and we want to estimate the mean $\mu = \mathbb{E}_P[X]$.

Formally, this involves constructing intervals $\{\mathcal{C}_n\}_{n \geq 1}$, adapted to the data, such that

$$\mathbb{P}_P(\forall n \geq 1 : \ \mu \in \mathcal{C}_n) \geq 1 - \alpha.$$

This is a level-$(1-\alpha)$ confidence sequence (CS) for μ



Contrast with fixed-time confidence intervals:

$$\underbrace{\mathbb{P}_P(\mu \in \mathcal{C}_n) \geq 1 - \alpha}_{\text{fixed } n} \qquad \text{vs.} \qquad \underbrace{\mathbb{P}_P(\forall n \geq 1 : \ \mu \in \mathcal{C}_n) \geq 1 - \alpha}_{\text{all times simultaneously}}.$$

The coverage guarantee holds for all times simultaneously; hence for all random stopping times as well.

Constructing CSs by inverting anytime-valid tests

For each candidate mean $m \in [0,1]$, test the point null $H_{0,m}: \ \mu = m$ using an evidence process $\{M_n(m)\}_{n\geq 1}$.

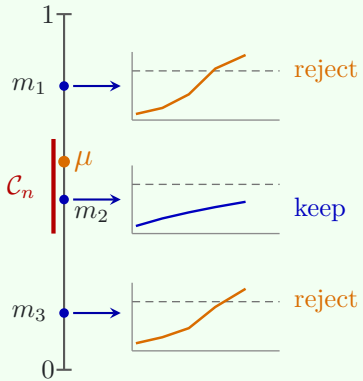
Invert the tests. Keep exactly those values that have not yet been rejected:

$$\mathcal{C}_n = \left\{ m : \ \max_{1\leq s\leq n} M_s(m) < \frac{1}{\alpha} \right\}.$$

At $m^\star = \mu$, we have by Ville's inequality,

$$\mathbb{P}_P(\exists n \geq 1 : \ m^\star \notin \mathcal{C}_n) = \mathbb{P}_P\left(\sup_{n\geq 1} M_n(m^\star) \geq \frac{1}{\alpha} \right) \leq \alpha.$$

We can also characterize the optimal size of $|\mathcal{C}_n|$ in terms of inverse $\mathsf{D}_{\mathsf{inf}}$ terms.



Modified Observation Model 1: Markov chains

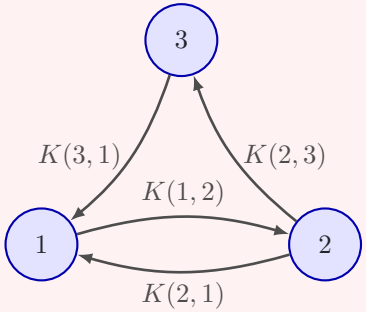
Consider a Markov chain on a finite state space $\mathcal{S} = \{1, \ldots, k\}$.

Let $K(\cdot, \cdot)$ denote the **unknown** transition matrix:

$$\mathbb{P}_K(X_t = y \mid \mathcal{F}_{t-1}) = K(X_{t-1}, y).$$

Testing problem

$$H_0 : \ K \in \mathcal{K}_0 \qquad \text{vs.} \qquad H_1 : \ K \notin \mathcal{K}_0.$$



Goal: Design a level-α power-one test τ for the above problem with small $\mathbb{E}_K[\tau]$.

The lower bound for Markov chains

Consider an ergodic alternative K with stationary distribution π_K

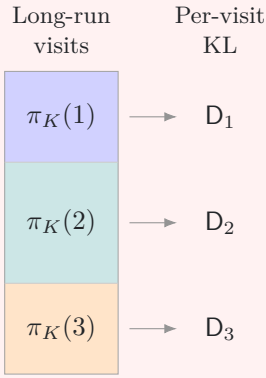
The relevant information quantity is

$$\mathsf{D}_{\inf}^{\mathrm{MC}}(K \parallel \mathcal{K}_0) := \inf_{L \in \mathcal{K}_0} \sum_{x \in \mathcal{S}} \pi_K(x)\, \mathsf{D}\big(K(x,\cdot) \parallel L(x,\cdot)\big).$$

Lower bound. Every level-α, power-one test satisfies

$$\mathbb{E}_K[\tau_\alpha] \geq \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}^{\mathrm{MC}}(K \parallel \mathcal{K}_0)} - O_K(1).$$

Proof: Combination of DPI and a modified Wald's identity.



Universal inference for Markov Chains

Finite-state Markov UI: Replace symbol pmfs in the i.i.d. case by conditional transition probabilities

$$M_t^{\mathrm{MC}} = \frac{\prod_{i=2}^{t} \widehat{K}_i(X_i \mid X_{i-1})}{\sup_{L \in \mathcal{K}_0} \prod_{i=2}^{t} L(X_{i-1}, X_i)} = \frac{\text{Mixture of Conditional Likelihoods}}{\text{Maximum Null Conditional Likelihood}}.$$

Stopping time: $\tau_\alpha = \inf\{t \geq 2 : M_t^{\mathrm{MC}} \geq 1/\alpha\}$

Unchanged: Nullwise domination + Ville still guarantee anytime validity.

New tools: Universal predictors for Markov sources; Markov concentration; modified Wald's identity.

Under suitable conditions, $\mathbb{E}_K[\tau_\alpha] \sim \log(1/\alpha)/\mathsf{D}_{\inf}^{\mathrm{MC}}(K \parallel \mathcal{K}_0)$.

Modified objective 1: Horizon-aware testing

So far, we have focused on minimizing $\mathbb{E}_P[\tau_\alpha]$.

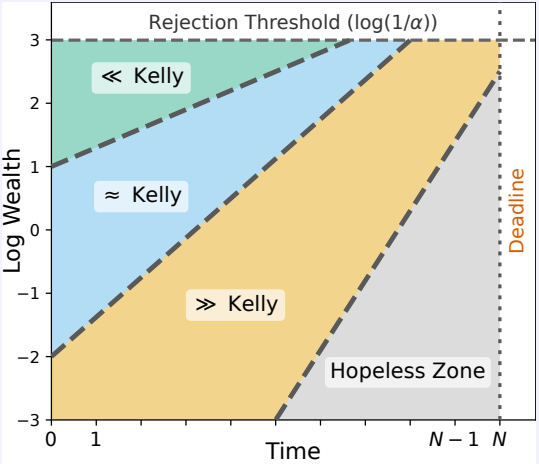
What if we have a **strict deadline** N?

Horizon-aware objective.

$$\begin{aligned} &\max \mathbb{P}_P(\tau_\alpha \le N), \\ \text{s.t.} \quad &\sup_{Q \in \mathcal{P}_0} \mathbb{P}_Q(\tau_\alpha \le N) \le \alpha. \end{aligned}$$

For the betting test, it reduces to

$$\max_{\{u_t\}_{t=1}^N} \mathbb{P}_P\left(\max_{1 \le t \le N} \prod_{i=1}^{t} (1 - u_i + u_i \frac{X_i}{m}) \ge \frac{1}{\alpha} \right).$$



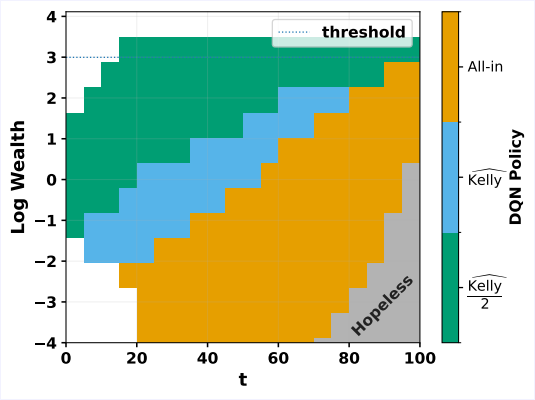
Horizon-aware betting

Finite-horizon control. For a fixed alternative P, the state is $(t, \log W_t)$; the action is the next feasible bet.

Theory. Under suitable conditions:

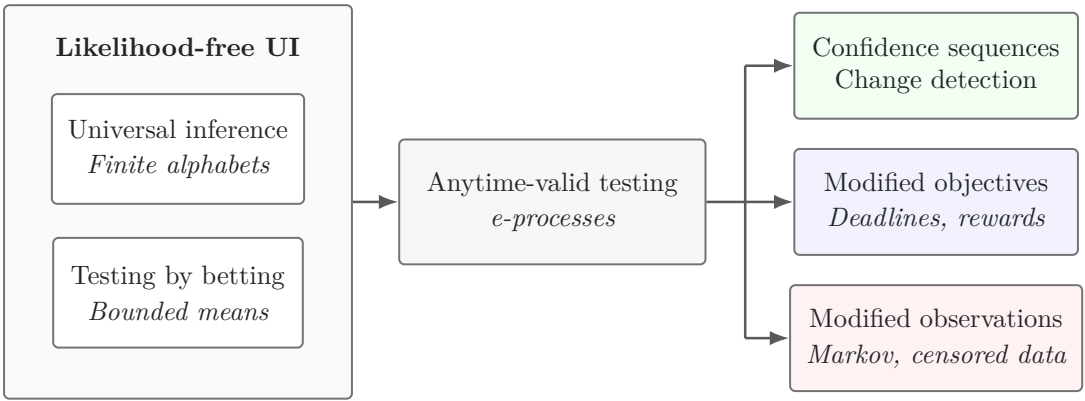
- **Ahead:** less aggressive than Kelly.
- **On schedule:** approximately Kelly.
- **Behind:** more aggressive than Kelly.

Unknown P. A Deep Q-Network (DQN) can learn to select bets using statistics of past observations.



Learned DQN actions.

Summary so far



Some Future Directions

Beyond first-order optimality

Our benchmark: for every fixed alternative P,

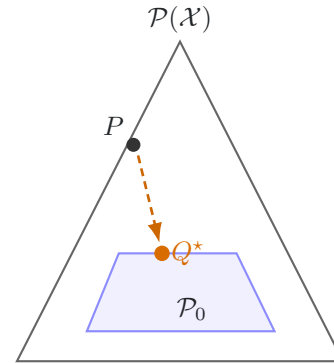
$$\mathbb{E}_P[\tau_\alpha] \sim \frac{\log(1/\alpha)}{\mathsf{D}_{\inf}(P \parallel \mathcal{P}_0)} \qquad (\alpha \downarrow 0).$$

Second-order behavior.

How does the stopping time fluctuate around its leading term?

Time-uniform local detection boundary.

What are the closest alternatives that can be reliably detected by a single anytime-valid test across all $n \in \mathbb{N}$?



Active anytime-valid inference

Multiple data sources. Suppose K sources are governed by a common hypothesis.

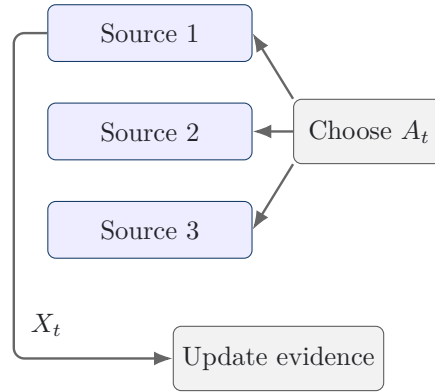
At each time t, the analyst chooses

$$A_t \in \{1, \ldots, K\}, \qquad X_t \sim P_{A_t},$$

using the observations collected so far.

Different sources may provide different amounts of information.

Question. What is the optimal joint policy for *sensing and stopping*?

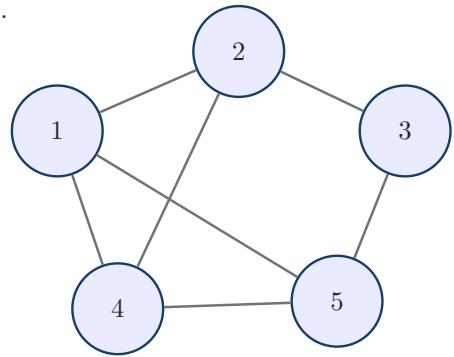


Distributed anytime-valid inference

Distributed observations: multiple agents collect local data, satisfying a common underlying hypothesis.

Each agent maintains local evidence $E_t^{(i)}$ and communicates with its neighbors (or a central server).

Question: How can a network aggregate evidence with both validity and optimal information growth?

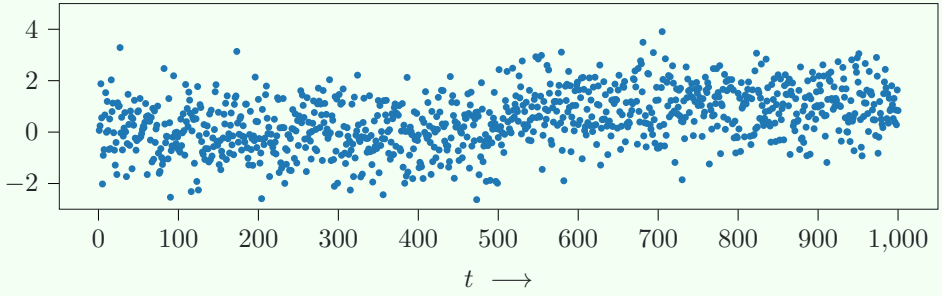


Thank you.

Modified Task 2: Sequential change detection

Suppose the distribution of the observations changes at an unknown time ν:

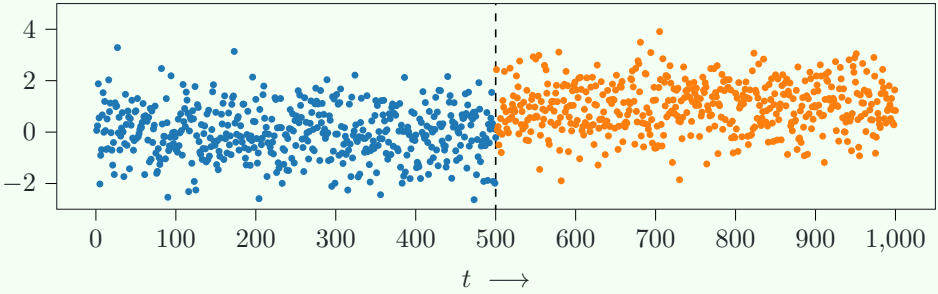
$$X_t \sim \begin{cases} P_{\text{pre}}, & t \leq \nu, \\ P_{\text{post}} \neq P_{\text{pre}}, & t > \nu. \end{cases}$$



Modified Task 2: Sequential change detection

Suppose the distribution of the observations changes at an unknown time ν:

$$X_t \sim \begin{cases} P_{\text{pre}}, & t \le \nu, \\ P_{\text{post}} \neq P_{\text{pre}}, & t > \nu. \end{cases}$$

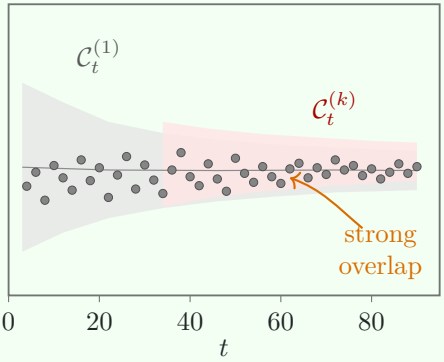


Goal: Detect the change quickly while keeping false alarms rare.

Change detection using confidence sequences

1. At each round k, start a new CS $\{\mathcal{C}_t^{(k)} : t \geq k\}$.
2. At time t, compare all active confidence sequences $\mathcal{C}_t^{(1)}, \mathcal{C}_t^{(2)}, \ldots, \mathcal{C}_t^{(t)}$.
3. Stop as soon as they *disagree*, i.e.

$$\tau = \inf\Big\{t \geq 1 : \bigcap_{k=1}^{t} \mathcal{C}_t^{(k)} = \varnothing\Big\}.$$

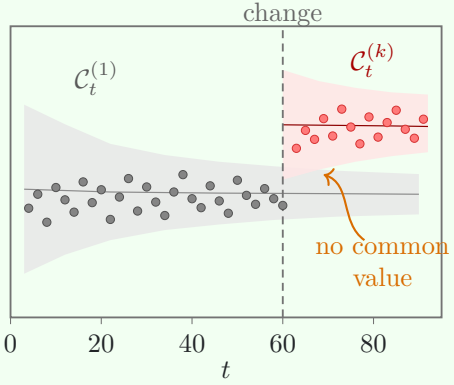


- When $\nu = \infty$, we have $\mathbb{E}_\infty[\tau] \geq \frac{1}{\alpha}$.
- When $\nu < \infty$, detection delay depends on quality of the CSs used.

Change detection using confidence sequences

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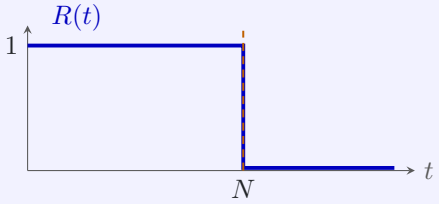
- When $\nu = \infty$, we have $\mathbb{E}_\infty[\tau] \geq \frac{1}{\alpha}$.
- When $\nu < \infty$, detection delay depends on quality of the CSs used.

Modified objective 2: Time-sensitive testing

A deadline is one way to reward early detection. More generally, assign a nonnegative, nonincreasing reward $R(t)$ to rejection at time t.

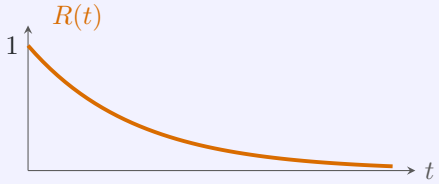
$$\sup_{\tau_\alpha \in \mathcal{T}_\alpha} \mathbb{E}_P[R(\tau_\alpha)], \qquad R(\infty) = 0,$$

Hard deadline



$R(t) = \mathbf{1}\{t \le N\}$ gives $\mathbb{P}_P(\tau_\alpha \le N)$.

Discounted discovery



$R(t) = e^{-ct}$ gives $\mathbb{E}_P[e^{-c\tau_\alpha}]$.

Rényi divergences and reward decay

Converse (ongoing work). For a finite-alphabet alternative $P \notin \mathcal{P}_0$, let $\mathcal{I}_r(P) := \inf_{Q \in \mathcal{P}_0} D_r(P \,\|\, Q)$.
Write $\Phi_P(\eta) := \eta \mathcal{I}_{1/(1-\eta)}(P)$, where $0 < \eta < 1$. Every level-α test satisfies

$$\mathbb{E}_P[R(\tau_\alpha)] \;\le\; \inf_{0<\eta<1} \left\{ \alpha^\eta \sup_{n \ge 1} R(n) e^{n\Phi_P(\eta)} \right\}.$$

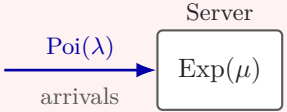
Three regimes, depending on how rapidly the reward decays:

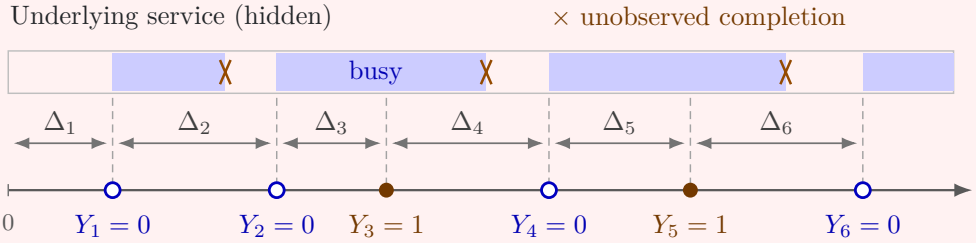
Subexponential $R(n) = e^{-o(n)}$	KL / typical stopping time $\log(1/\alpha)/\mathsf{D}_{\inf}(P \,\|\, \mathcal{P}_0)$; a separate argument is needed.
Exponential $R(n) = e^{-cn}$	For $0 < c < \mathcal{I}_\infty(P)$, the Rényi exponent η_c solves $\Phi_P(\eta_c) = c$.
Superexponential $R(n) \lesssim e^{-cn}$	If $c \ge \mathcal{I}_\infty(P)$, the reward is $O(\alpha)$; randomized early rejection attains this order.

Here $\mathcal{I}_\infty(P) := \inf_{Q \in \mathcal{P}_0} D_\infty(P \,\|\, Q)$.

Modified Observations 2: Censoring

- A single server with service times $S \sim \text{Exp}(\mu)$.
- Jobs arrive as a Poisson process (rate λ).
- An arriving job is admitted whenever the server is idle.





Observations: inter-arrival times $\{\Delta_t : t \geq 1\}$ and busy/idle indicators $\{Y_t : t \geq 1\}$.

Goal: For some known $\mu^\star > 0$, decide between $H_0 : \ \mu \geq \mu^\star$ vs. $H_1 : \ \mu < \mu^\star$.

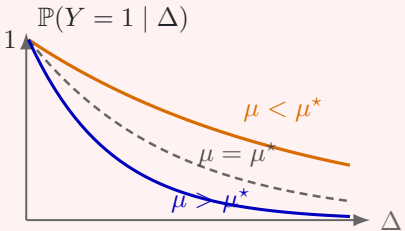
Anytime-Valid Test From Occupancy Indicators

Challenge: We want to test a property of service times $\{S_t : t \geq 1\}$, but we don't directly observe them!

Under the Poisson-arrival assumption: $\Delta_t \sim \mathrm{Exp}(\lambda)$.

Under the additional Exponential service-time assumption,

$$Y_t \mid (\Delta_t, \mathcal{F}_{t-1}) \sim \mathrm{Ber}(e^{-\mu \Delta_t}).$$



An instantiation of the likelihood-based universal inference procedure gives, for fixed $\mu < \mu^\star$,

$$\mathbb{E}_{\mu,\lambda}[\tau_\alpha] \sim \frac{\log(1/\alpha)}{I_{\mu,\lambda}}, \quad \text{where} \quad I_{\mu,\lambda} = \mathbb{E}_\Delta\left[\mathsf{d}\big(e^{-\mu\Delta}, e^{-\mu^\star \Delta}\big)\right],$$

Similar ideas extend to more general service time distributions.